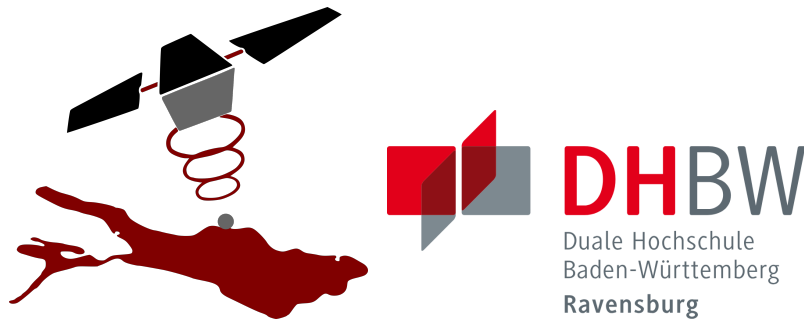


Project Plan

for

Development of a telemetry data link for the
ERWIN cubesat



Project:	HW/SW Projekt TSL - 0004
Date:	June 11, 2026
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Description:	Document to define the overall structure of the Project

Revision History

Name	Date	Description	Version
Lukas Reil & Nico Tunkowski	29.10.2025	Document aggregation	Draft
Lukas Reil & Nico Tunkowski	30.10.2025	Requirements refining	v1.0
Nico Tunkowski	30.10.2025	Weekly Mail content	v1.1
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1 References

Standard Number	Title
CCSDS 133.0-B-2	Space Packet Protocol
CCSDS 132.0-B-3	TM Space Data Link Protocol
CCSDS 131.0-B-5	TM Synchronization and Channel Coding
ECSS-M-ST-10C Rev. 1	Space project management
SeeSat-TMTC-101	VHDL Development Guideline
SeeSat-TMTC-102	C++ Coding Guidelines
SeeSat-TMTC-103	Development Process
SeeSat-TMTC-104	Git Standardization and Workflow

2 Abbreviations

AI	Action Item
ASM	Attached Synchronization Marker
AXI	Advanced eXtensible Interface
BVSR	Bundesverband studentischer Raumfahrt
CCSDS	Consultative Committee for Space Data Systems
CRC	Cyclic Redundancy Check
DLRK	Deutscher Luft- und Raumfahrtkongress
ECSS	European Cooperation for Space Standardization
ERWIN	Emission of Radiation based Wildfire Investigation
ETH	Ethernet
FECF	Frame Error Control Field
MVP	Minimum Viable Product
OBC	OnBoard Computer
OCF	Operational Control Field
OSI	Open Systems Interconnection
PL	Programmable Logic
PS	Processing System
R/S	Reed Solomon
SoC	System on a chip
SDLS	Space Data Link Security
TCP	Transmission Control Protocol
TM	Telemetry

3 High Level Requirements

3.1 System

SYS-REQ-1 Space Packets (except idle space packets) that are send to the System on the encoder input shall be outputted on the

decoder output, these packets shall be identical

- SYS-REQ-2** Space Packets shall be delivered on Ground with the same bit order as in Space provided
- SYS-REQ-3** The Pseudo Random Number generator shall use the generator polynomial $h(x) = x^{17} + x^{14} + 1$ (ref. to CCSDS 131.0-B-5 10.4.1)
- SYS-REQ-4** The Pseudo Random Number generator shall use the starting Condition 1000111000111000 (ref. to CCSDS 131.0-B-5 10.4.3)
- SYS-REQ-5** The Reed Solomon (R/S) shall use $F(x) = x^8 + x^7 + x^2 + x + 1$ over a galois field of GF(2) (ref. to CCSDS 131.0-B-5 4.3.3)
- SYS-REQ-6** The R/S Encoder shall have a error correction capability of 16 symbols per R/S block ($E = 16$)
- SYS-REQ-7** The Convolutional code shall use $G1 = 1111001$ and $G2 = 1011011$ as Connection vectors
- SYS-REQ-8** The Convolutional code shall use a 50% Code rate
- SYS-REQ-9** There shall be NO interleaving used in the Reed Solomon generator

3.2 Space

- SPC-REQ-1** The system shall not loose space packets, except idle space packets, in the processing chain
- SPC-REQ-2** The system shall be able to produce at least 2 Mbit/second of telemetry (TM) data, on the output of the 131 encoder
- SPC-REQ-3** The system shall interface with the Processing System (PS) with an advanced extensible interface (AXI)
- SPC-REQ-4** The AXI interface between PS and system shall be clocked to fulfill the data throughput requirements

SPC-REQ-5 The system shall interface with the antenna system using an AXI interface

SPC-REQ-6 The system shall be implemented on a ZYNQ 7000 (PYNQ Z2) system on a chip (SoC)

3.2.1 CCSDS-132.0-B-3

SPC-REQ-7 The TM transfer frame length shall be exactly 2046 octets (SeeSat-1-CCSDS-Stack)

SPC-REQ-8 The system shall be CCSDS-132.0-B-3 compliant

SPC-REQ-9 The operational control field (OCF) should be implemented

SPC-REQ-10 The OCF should be configurable on synthesis

SPC-REQ-11 The frame error control field (FECF) should be implemented

SPC-REQ-12 The FECF should be configurable on synthesis

SPC-REQ-13 The system shall be able to pack partial space packets into TM transfer frames

SPC-REQ-14 The system shall be able to calculate the payload trailer of the TM transfer frames without the space data link security (SDLS) attachment

SPC-REQ-15 The system shall be able to send data to the TM SYNCHRONIZATION AND CHANNEL CODING layer (CCSDS 131.0-B-5)

SPC-REQ-16 The system shall be able to receive data from multiple sources (Virtual Channels)

SPC-REQ-17 The system should implement multiple virtual channels

SPC-REQ-18 The system should implement multiple master channels

SPC-REQ-19 The system should implement a secondary frame header

SPC-REQ-20 The secondary frame header should be configurable

3.2.2 CCSDS-131.0-B-5

- SPC-REQ-21** The system shall use concatenated coding scheme
- SPC-REQ-22** The system shall use Reed Solomon coding for outer codes
- SPC-REQ-23** The system shall use pseudo randomization
- SPC-REQ-24** The system shall attach a 32-Bit attached synchronization markers (ASM) to the data stream
- SPC-REQ-25** The system shall use convolutional coding for inner coding
- SPC-REQ-26** The system shall expose its configuration, in accordance with configuration management

3.3 Ground

- GND-REQ-1** The system shall be able to indicate to the user when a packet cannot be corrected
- GND-REQ-2** The system shall be able to correct errors that are correctable by the algorithms used
- GND-REQ-3** The system shall interface with further systems via TCP packets
- GND-REQ-4** Transmission of correct(ed) packets to further Mission Control systems shall only include the binary data of the space packet
- GND-REQ-5** The ground system shall be implemented on a ZYNQ 7000 (PYNQ Z2) SoC

3.3.1 CCSDS_131.0-B-5

- GND-REQ-6** The system shall be connected to the receiving antenna via Ethernet/TCP or AXI Stream.

-
- GND-REQ-7** The system shall receive TM data from the antenna as raw bits
- GND-REQ-8** The system shall decode these raw bits using an appropriate decoding algorithm for convolutional coding
- GND-REQ-9** The system shall subtract the injected pseudo random values from the data returned by the asm decoder
- GND-REQ-10** The ground system shall verify the received bit count using the transmitted Attached Synchronization Marker (ASM)
- GND-REQ-11** The system shall decode the Reed Solomon codes
- GND-REQ-12** The data generated by this layer (CCSDS 131) shall be transmitted to CCSDS 132

3.3.2 CCSDS_132.0-B-3

- GND-REQ-13** The system shall extract the space data packets
- GND-REQ-14** The extracted space data packets shall be send via TCP to the next systems
- GND-REQ-15** The system shall de-multiplex the space packet to its corresponding Virtual- and Master channel
- GND-REQ-16** The system should implement the decoding of the FECF
- GND-REQ-17** The FECF should be configurable on synthesis
- GND-REQ-18** The system should implement the decoding of the OCF
- GND-REQ-19** The OCF should be configurable on synthesis
- GND-REQ-20** The system should implement a secondary frame header
- GND-REQ-21** The secondary frame header should be configurable
- GND-REQ-22** If data is not correctable by the CCSDS 131, uncorrected data shall still be received by CCSDS 132 and sent out to higher protocol layers if possible

3.4 Non Functional

NFUNC-REQ-1 The systems program code and hardware configuration shall follow a clear and defined standard (See VHDL Standard - SeeSat-TMTC-101)

NFUNC-REQ-2 The systems program code and hardware configuration shall be written in an extendable and maintainable way

4 Project Time Management

The project timeline is divided into the product's development sections and the mandatory meetings and deadlines provided by the customer. The different development phases and deadlines can be seen in the Gantt diagram of the project timeline 4.1.

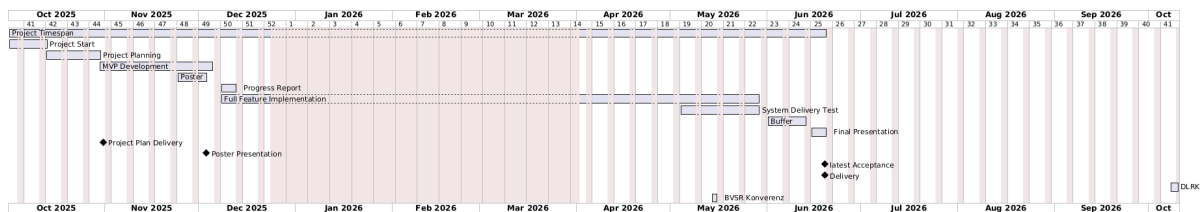


Figure 4.1: Gantt Timeline

4.1 Timeline

As can be seen in the Gantt diagram, the project has a total length of 38 weeks, of which 24 weeks or 120 days are available for development. These available development days are grouped into phases. These will now be explained in detail.

4.1.1 Project Planning

During this phase, the team meets to review the tasks involved in the project. The project development plan and other key documents are also written during this phase. Only high-level requirements are written on the technical side.

4.1.2 Minimum Viable Product (MVP) Development

In this phase, the team will begin developing critical systems and functionality. The goal is not to develop the complete scope of the product, but rather to create something that can be presented to users and demonstrate that the team's decisions are functioning as intended. If the customer has any changes they would like to make, they can be incorporated at this point without causing any major delays.

4.1.3 Poster

During this phase, the team creates the poster to be displayed at the poster session. This phase runs in parallel with the development process and is not directly tied to any progress made during this process.

4.1.4 Progress Report

During this phase, the team will meet with the customer to discuss whether the work completed so far meets their standards and expectations. Changes to the project scope are still possible at this stage, but depending on the change request, they may result in a significant time penalty.

4.1.5 Full Feature Implementation

In this phase, the team will develop the full feature set that was not implemented in the MVP phase. Changes to the scope or architecture at this stage will incur significant time penalties, which will most likely result in missed deadlines or project failure.

4.1.6 System Delivery Test

In this phase, the product is tested against the customer specifications and high-level requirements, according to the development process model. Lower-level tests and unit tests are carried out in parallel with the implementation of individual features and are therefore not listed in this timeline.

4.1.7 Buffer

The purpose of this phase is to act as a buffer, ensuring that missed deadlines or human error during the development phase do not impact the project delivery deadline.

4.1.8 Final Presentation

During this phase, the final product is presented to the customer for review. The date of this presentation is determined by the customer, who must give the development team at least two weeks' notice. If the customer does not provide a date for the final presentation, the development team will assume that it will occur on the last day of this phase.

4.2 Deadlines

The following sections describe the various deadlines relevant to the development team. These deadlines do not include those outside the project scope, such as the BVSR conference and the Deutscher Luft- und Raumfahrtkongress (DLRK).

4.2.1 Project Plan Delivery

By this deadline, the team will provide the project plan that was created during the planning phase. Any changes to the scope or features after this date will result in the generation of a new version of the project plan.

4.2.2 Poster Presentation

By this deadline, the team will present the poster created during the project poster phase. The deadline for deciding what can be included in the poster must be set before the poster phase begins.

4.2.3 Latest Acceptance

This deadline marks the latest time the customer will accept the project. The customer is expected to judge the project based on the most recent version of the agreed project plan document.

4.2.4 Delivery

This deadline marks the date by which the project must be completed. It also marks the end of the official project period. However, other work, such as writing a paper, can continue after this date.

4.2.5 Other Phases and Deadlines

The team intends to write a paper about the project's implementation. The timeline for writing and submitting the paper will be announced at a later date.

5 Scope

The scope of this project is bounded by CCSDS 132 and CCSDS 131. These standards define the ground and on board side of open systems interconnection (OSI) layer 2 (data link layer). Appropriate interfaces to all inbound and outbound systems will be provided, defined in the respective ICDs. Furthermore, an emulation of all relevant systems for a full hardware in the loop validation will be developed, if not already developed and supplied by SeeSat. As the relevant ICDs have not been

supplied at the time of writing, it is assumed that one instance of this system will be integrated into the OnBoard programmable logic (PL) for every physical TM channel.

6 Goals

The goal of the project is to develop the ground and on board side of OSI Layer 2 of ERWIN's TM chain. As already specified in the scope, the goal is to be compliant with CCSDS 132.0-B-3 and CCSDS 131.0-B-5.

6.1 CCSDS 132.0-B-3

The current design aims to implement full first header support. The secondary header is not going to be implemented. Likewise, neither the OCF nor the FECF parts of the trailer are going to be implemented. Insertion of idle data, to keep a consistent data rate, is part of the current goal. Only one master channel, but multiple virtual channels will be supported, if support for multiple channels is required by ERWIN's system design.

6.2 CCSDS 131.0-B-5

As the FECF is not going to be implemented the following measures for error correction are going to be developed in the lower layer: Reed Solomon will form the inner error correction method, with the following parameters: $E = 16$, $I = 1$ and virtual fill disabled. Furthermore, the outer error correction method will consist of a convolutional code of code rate $r = 50\%$. Between these layers the suggested randomization, with polynomial of degree 17 and the required ASM field will also be implemented.

6.3 Test System

Additionally a full hardware in the loop test system is planned. For that purpose 3 additional systems are going to be implemented.

6.3.1 OBC Emulator

A stub On-Board-Computer will be developed, to allow sending fake space packets from the processing system (PS) of the target SoC to the CCSDS 132.0-B-3 layer, implemented on the PL.

6.3.2 AXI to ETH Interface

To allow routing fully processed TM frames out of the OnBoard SoC, an interface will be developed, to send out the frames via an Ethernet interface, in the same format, as the ground system requires them.

6.3.3 Control Software

Finally a control software will be developed, which will manage generating the fake space packets for the OBC emulator and also receive the final unpacked space packets decoded by the ground system. Therefore, the control software will be able to monitor packet loss, latency (to some degree) and throughput of the final system. Additionally the software will be able to intercept packets generated by the AXI to ETH interface, to simulate upsets in the transmission to also test the error correction's performance.

6.4 Additional Goals

If all previously specified goals are met, additional goals will be implemented, to accommodate a broader usage of the developed product. These additional features will focus on more CCSDS 131.0-B-5 features, to allow a mission specific configuration of data flow. Specifically these goals are the integration of a secondary header into the TM frame, as well as a cyclic redundancy check (CRC) checksum in the FECF. Lastly the OCF

will be made available, but will not be filled with meaningful data, as this is up to the mission specification.

7 Development Process

The development team shall use a tailored V-Model as their development process model. The V-Model shall be subdivided into 4 horizontal layers:

1. User Layer
2. High Level Layer
3. Low Level Layer
4. Implementation Layer

The User Layer shall represent the SeeSat e.V. This instance is the final user of the product, this layer defines the use cases and acceptance tests. The High Level Layer shall represent system requirements and system tests. The Low Level Layer shall define unit tests and requirements that can be directly mapped to the Implementation Layer. In addition to the classical V-Model an iterative approach is used in the lower two layers of the V-Model, to allow for an iterative development. This means that each subsystem of the project may be advanced at its own respective pace and if a logical mistake is noted in the development or testing of a specific module one may change the respective low level requirements. A main target of this iterative approach is to develop working prototypes to make sure that the final product is in a working and deliverable state after every sprint.

Git shall be used for versioning and work distribution. The team may also use GitLab's CI/CD pipeline for automatic testing and generating artifacts from source, e.g. build files or documents from L^AT_EX. For proper usage of Git a standard has been defined by the development team that can be found in SeeSat-TMTC-104: Git Standardization and Workflow. If

the Customer is able to provide a Valispace environment the development team should use Valispace as a requirements management Tool. At the time of writing there is no such environment (see. Risk Management 8) If the development team draws diagrams these diagrams shall be drawn in the free online tool draw.io or in the textual UML tool PlantUML.

A detailed process definition of the Development Process used in this project can be found in SeeSat-TMTC-103: Development Process.

8 Risk Management

As part of this project, the team conducted a comprehensive risk analysis. Each of the risks that arose were considered in terms of probability of occurrence, severity of consequences and possible reactions. These risks are also recorded in a separate document, which is to be extended if new risks are identified.

8.1 Identified Risks

8.1.1 Lack of experience regarding Project size

One of the identified risks is the teams missing experience working on and managing a project of this scope. It is clear that at some point this can result in several issues caused by miscalculation of time needed for specific tasks and/or mistakes being made during planning phase of the project. While it is necessary to allow for mistakes to be made in order to learn, additional time buffers are needed to prevent a delayed project delivery.

Problems like this are highly likely to occur and can have a catastrophic effect on the projects timeline if not accounted for.

8.1.2 Communication with Customer

The experience of other projects in association with the customer SeeSat e.v. has shown, that in some cases information that is necessary for the project isn't always clearly defined and provided. Additionally it is clear, that the persons responsible for this project have other accountabilities and therefore may not always be able to answer in time. Of course this can result in delayed information and have effects on the projects timeline.

To prevent delays regarding the projects timeline, critical Information needs to be requested at least 1, better 2 weeks before it's needed.

8.1.3 Unclear Interfaces to other SeeSat Tech

On the space side it is unclear how the developed components shall interface with the OBC or UHF Modem for example. Of course this problem doesn't directly affect the project but may cause issues in the integration state of ERWIN. Due to the team having interests in the success of the Satellite we decided to mark this down as an additional risk. It may be a preliminary solution to keep the previously mentioned interfaces as modular as possible to allow for easy adaption later on.

8.1.4 Previous work

This project is the continuation of 2 previous projects, whose task it was to develop the the Telemetry Front End on the Ground and on Space side. As they were unable to completely finish their projects it is our task to develop one function module for the ground and the space side. It may need more work to get acquainted with the previously developed code that to entirely rework it. This especially opposes a risk as the previously developed space segment does not include all necessary features, so that it maybe needs to be developed from ground up.

If this gets clear after substantial work was put into the group familiarizing itself with the already developed VHDL Code elements, it may threaten the project timeline.

To prevent this from occurring it is necessary to start by checking if the VHDL structure can be extended by the necessary encoding structures.

8.1.5 Unknown Hardware

Due to the team members not being familiar with the provided hardware, it is unclear if the structures and interfaces that have been decided on so far are feasible. The team has decided that this is a more minor risk as the majority of the developed structures is more or less hardware independent. If a member however runs into issues regarding this, it is important to report it to the other members as soon as possible to prevent the timeline being affected by this.

8.1.6 CCSDS 131.0-B-5

The implementation of the CCSDS 131.0-B-5 imposes several risks:

Firstly especially the decoding actions of Reed Solomon and Convolutional Code are challenging on the mathematical side. It is unclear if any team member is mathematically skilled enough to be able to implement the decoding action in the time necessary. It is rather unlikely that no one will be able to do this at all but the needed time can cause problems. If however no one manages to implement said functionalities this can have catastrophic effects on the projects execution, as other solutions need to be found in that case.

Team Members have also mentioned that the encoding process may need more than one clock cycle per encoded Bit which may cause timing issues before and after the encoding stages. The development team needs to be aware that this may cause issues and should be open to design changes in the affected parts.

8.1.7 Knowledge Distribution and Personal Availability

As the project scope is quite comprehensive and the team size of 5 members is relatively small compared to it, it is quite difficult that for each field of knowledge there are at least 2 persons able to support others in. This can lead to issues if a team member is not available due i.e. illness or other personal reasons. As long as the non availability is less than a week the consequences of this are rather small, if however a team member is away for more than a week it can have noticeable effects on the projects

development timeline.

8.1.8 Time allocated by other Projects

It is obvious that this project isn't the only project of the team members this semester and the team has decided that the "Studienarbeit" should be prioritized. This causes a Risk for the project as it is not guaranteed that everyone will be able to put in the necessary work at all times.

To prevent this from having substantial effects on the projects outcome it should be considered during the planning stages of the project and when creating an overview of the projects milestones.

9 Communication

In this chapter the communication between the customer (SeeSat e.V.) and Supplier (TSL23 TM Group) is described. The methods of information exchange are stated and cyclic intervals defined.

9.1 Reporting System

To inform the customer a reporting system via E Mail is established. The responsible on the supplier side writes a weekly E Mail containing information on project status with additional questions from the team. The questions part includes old not yet answered and new ones. The report includes a brief update on current and upcoming tasks.

9.2 Action Monitoring

The action monitoring is split into two categories. The first consists of monitoring general tasks and the second is the management of code writing. General tasks are tracked with action items (AI) who are conducted after each meeting. An AI is used to communicate the tasks each team

member does at the moment. Therefore each AI is assigned a person. After working on a task the work is reviewed by other team members internally before the AIs deadline. Code related work items are monitored on GitLab using a ticket system further explained in the development process document.

9.3 Customer-Supplier Meetings

There are no regular meetings planed with the customer but there are regular update E Mails send by the supplier.

9.4 Meeting Documentation

In each meeting one person is responsible to write notes that document the most important content discussed and to save it for later use. The responsible is chosen before the start of every meeting. The document will be saved to the corresponding meetings folder on the SeeSat Nextcloud in the meetings folder followed by the date the meeting was held.

9.5 Responsibility

In the weekly team meeting the content for each update mail is conducted. Afterwards additions like new risen questions and not yet answered questions (chapter Reporting System) are written by the communication responsible Nico. The E Mail is shared with the team members after its first iteration and additions or changes can be added to form a second iteration. After a “yes” from every team member the E Mail is send to the customer contacts Marcel and Stefan on the end of the week. Responses from the customer are shared as soon as possible with the team.

9.6 Internal Team Communication

The primary online communication tool used is a private discord channel. All event times are written there to be accepted or declined by every team

member to centrally collect and inform every person. Status updates and requests for reviews are written in the chat. This allows every member to see the current problems or help each other.

9.7 Document Exchange

In the project lifetime many documents will be created. To exchange documents with the customer for explicit review a folder structure on the SeeSat Nextcloud created by the customer. Therefore a folder named "From SeeSat" is used to receive documents and a folder named "To SeeSat" to send documents. It is expected that both sides will read its corresponding received documents and communicate any changes that is wished to be made. Otherwise the content will be understood as acknowledged.

10 Project Breakdown Structure

For better understanding of the projects composition, the following explains its inner Structure and the different components its made up of.

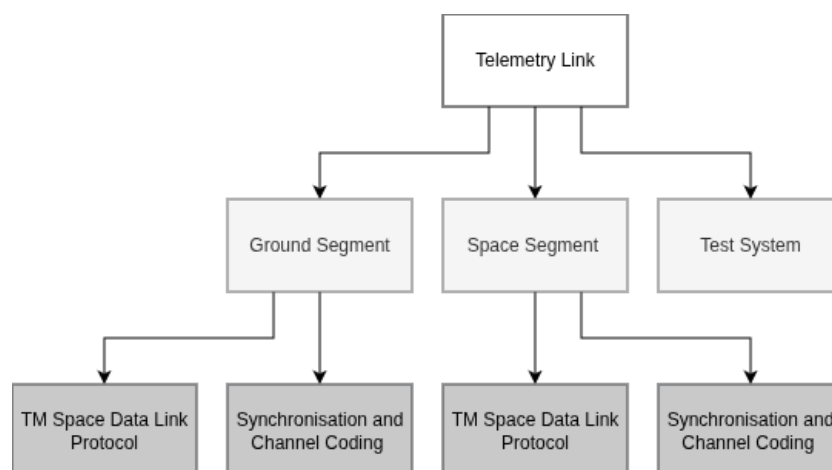


Figure 10.1: Project Structure

The entire project can firstly be divided into 3 main subsystems shown by figure 10.1. The TM Segment on Ground and its counterpart referred to as Space Segment, as well the necessary Test System for Hardware in the Loop testing planned towards the end of project. Both Ground and Space Segment are further divided into the implementation of the TM Space Data Link Protocol described in CCSDS 132.0-B-3 and the Synchronization and Channel Coding of CCSDS 131.0-B-5.